Absorbance Spectra and Standard Curves of Food Colorings

Chemistry 250L – Experiment 2: Spectrophotometers and Standard Curves

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Abstract

Beer's law was used to build a quantitative spectrophotometric method for four commercial food colorings (blue, green, yellow, red) of unknown composition. For each dye a diluted stock solution was scanned to find the wavelength of maximum absorbance (λ_{max}), and a five-point two-fold serial dilution was measured at that wavelength. The measured λ_{max} values were 629.5 nm (blue, two trials), 628.1 nm (green), 428.9 nm (yellow) and 522.9 nm (red). The standard curves were linear with R^2 between 0.9975 and 0.9996 (all above the 0.97 rubric criterion). Sensitivities (slopes) were 1.50, 1.72, 5.92, 0.605 and 2.39 absorbance units per µL/mL for blue 1, blue 2, green, yellow and red. The maximum absorbance was below 1 for four of the five curves; the yellow top standard read 1.032. The two blue slopes differ by about 15 %, which is more than their standard errors explain. Positive intercepts (up to 0.056) were found for all curves and were statistically significant for blue 2, yellow and red.

1 Introduction

In a quality-control setting, the color of a beverage must match its specification from batch to batch. Absorption spectrophotometry offers a fast way to check this, because the absorbance of a dye solution at a fixed wavelength is proportional to its concentration. This experiment models that task. The scenario is that a shipment of concentrated red, blue, green and yellow food coloring has lost its labels. The analyst must (1) choose the best wavelength to monitor each colorant from its absorbance spectrum, and (2) build a standard curve that can later be used to determine the concentration of the colorant in an unknown sample. The experiment also trains handling of concentrated, staining solutions (the food coloring simulates a hazardous substance) and use of a simple spectrophotometer.

2 Background and Theory

2.1 Food Colorings

A colorant appears colored because it absorbs some wavelengths of visible light and transmits or reflects the others, so the observed color is related to the wavelengths that are not strongly absorbed. Table 1 lists the FD&C dyes from the course reading, with their reference λ_{max}. A blue dye absorbs orange-red light near 630 nm, a red dye absorbs green light near 500 nm–530 nm, and a yellow dye absorbs violet-blue light near 430 nm.

Table 1: Reference values for FD&C dyes (course reading, "The Chemistry of Food Coloring").

Dye	Common name	Color	M (g/mol)	Reference $\lambda_{\max}$ (nm)
FD&C Blue No. 1	Brilliant Blue FCF	blue	793	625–630
FD&C Green No. 3	Fast Green FCF	green	809	625
FD&C Red No. 3	Erythrosine	pink/red	898	525–530
FD&C Red No. 40	Allura Red AC	red	496	504
FD&C Yellow No. 5	Tartrazine	yellow	534	426
FD&C Yellow No. 6	Sunset Yellow FCF	orange	452	480

2.2 Beer's Law and the Choice of $\lambda_{\max}$

Absorbance A at a given wavelength obeys the Beer–Bouguer–Lambert law

$$A = \varepsilon \, \ell \, c, \tag{1}$$

where ε is the molar absorptivity (extinction coefficient), ℓ the path length (1 cm for the cuvettes used) and c the molar concentration. Measuring at $\lambda_{\max}$ gives the largest signal per unit concentration, so a small concentration change gives a larger absorbance change, and the effect of a small wavelength-setting error is smallest because the spectrum is flat at its peak. At high absorbance (above about 1) less light reaches the detector and the reading becomes less reliable, so concentrated samples should be diluted.

2.3 Standard Curve and Dilution

Because the concentration of each commercial coloring is unknown, the concentration axis is expressed as the volume of undiluted dye per volume of solution, in µL/mL. The standard curve is a plot of A versus c with the fit

$$A = m c + b, \tag{2}$$

where the slope m is the sensitivity of the method. In terms of Eq. (1), $m = \varepsilon \, \ell \, c_{\mathrm{dye}} \times 10^{-3}$, where c_{dye} is the molar concentration of the undiluted dye and 10^{-3} converts µL/mL to a volume fraction. The stock was prepared by dilution, and each standard was made by a two-fold serial dilution:

$$c_{\mathrm{stock}} = \frac{V_{\mathrm{dye}}}{V_{\mathrm{final}}}, \qquad c_{n+1} = \frac{c_n}{2}. \tag{3}$$

2.4 Statistics Used

For $n = 5$ standards the residual standard deviation, the standard errors of slope and intercept, and the 95 % confidence interval ($t_{0.975,3} = 3.182$) are

$$s_{y/x} = \sqrt{\frac{\sum (A_i - \hat{A}_i)^2}{n-2}}, \quad s_m = \frac{s_{y/x}}{\sqrt{S_{xx}}}, \quad s_b = s_{y/x}\sqrt{\frac{1}{n} + \frac{\bar{c}^2}{S_{xx}}}, \quad \mathrm{CI} = \pm t \, s. \tag{4}$$

The limits of detection and quantitation are estimated as

$$\mathrm{LOD} = \frac{3.3 \, s_{y/x}}{m}, \qquad \mathrm{LOQ} = \frac{10 \, s_{y/x}}{m}. \tag{5}$$

With only five points and no replicates these are rough estimates.

3 Objectives

1. Handle concentrated dye solutions without contaminating hands, clothing, equipment or workstation.
2. Use a simple spectrophotometer (SpectroVis) to measure an absorbance spectrum.
3. Determine the wavelength of maximum absorbance for each food coloring.
4. Prepare a stock solution with maximum absorbance of about 0.9–1.1, make a serial dilution, and generate a standard curve (absorbance versus concentration) for each dye.
5. Evaluate each curve for linearity ($R^2 > 0.97$), maximum absorbance (< 1), and reproducibility.

4 Hypotheses

(1) Each dye will show a single dominant visible absorption band at a wavelength consistent with its color (blue near 630 nm, red near 500 nm–530 nm, yellow near 430 nm). (2) Absorbance at $\lambda_{\max}$ will be linear in concentration over the range of the standards (Beer's law), with $R^2 > 0.97$. (3) Two independent blue trials should give the same slope within experimental error.

5 Materials and Equipment

Food coloring (red, blue, green, yellow), distilled water, glass cuvettes (1 cm path), SpectroVis spectrophotometer, micropipettes and tips, volumetric containers for the stock and standards, labels, gloves, safety goggles, paper towels.

6 Safety

The RAMP approach was used. **Recognize:** some food colorings are toxic at high concentration and with repeated use, and concentrated dye stains. **Assess:** the main risks are ingestion and skin or clothing staining; no flammable or corrosive materials were used. **Minimize:** safety goggles and gloves were worn and absorbent paper was placed under the work area. **Prepare:** spills were wiped up immediately with paper towels, and the eyewash and hand-wash stations were known. Calibration and dilution water were disposed of in the sink as directed, and the work area was wiped down.

7 Methods

1. **Workstation.** Goggles and gloves were put on, the bench was covered, and the cuvettes, pipettes, distilled water, labels and notebook were gathered. Every container was labeled with dye name, trial number, dilution and date, and glassware was inspected for cracks and residue.
2. **Absorbance spectrum.** A diluted solution of one dye was prepared so that the spectrum was on scale. A clean cuvette was filled about $\frac{2}{3}$ full with distilled water, the clear faces were wiped and aligned with the light path, and the instrument was blanked in spectrum mode. The diluted dye was scanned across the visible range and $\lambda_{\max}$ and its absorbance were recorded.

3. **Stock solution.** The concentrated dye was diluted so that the maximum absorbance was about 0.9–1.1 (if above ≈ 1.0, a more dilute solution was prepared and the measurement repeated). The volumes used are in Table 2.

4. **Serial standards.** A two-fold dilution scheme was used: each standard was half the concentration of the previous one (relative $1.00, 0.50, 0.25, 0.125, 0.0625$ of the stock). Distilled water was added before dye where possible, each standard was mixed thoroughly with a clean tip, and the exact concentrations were recorded.

5. **Measurement.** The wavelength was set to the measured $\lambda_{\max}$, cuvettes were checked for scratches and bubbles, the instrument was zeroed on the distilled-water blank, and the standards were measured from most to least concentrated with the clear faces wiped and the cuvette in the same orientation each time.

6. **Standard curve.** Concentration was plotted on the x-axis and absorbance on the y-axis as a scatter plot with a linear trendline, equation and R^2. Steps 2–6 were repeated for each dye, and blue was repeated for a fifth trial.

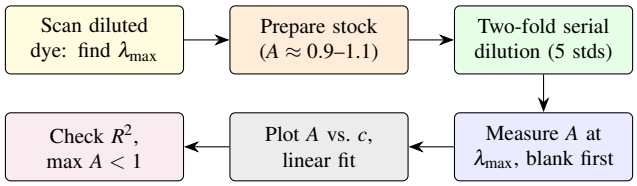


Figure 1: Workflow for each dye.

8 Results

8.1 Wavelength of Maximum Absorbance

Table 2 gives the measured $\lambda_{\max}$ and the stock preparation for each trial. The notebook records the $\lambda_{\max}$ and the absorbance of the diluted stock, but not the spectra themselves.

Table 2: Stock solutions and measured $\lambda_{\max}$.

Trial	Dye volume / final volume	Stock conc. (µL/mL)	$\lambda_{\max}$ (nm)	Top standard A
Blue 1	10 µL / 20 mL	0.500	629.5	0.774
Blue 2	5 µL / 10 mL	0.500	629.5	0.885
Green	3 µL / 20 mL	0.150	628.1	0.901
Yellow	20 µL / 12 mL	1.667	428.9	1.032
Red	8 µL / 20 mL	0.400	522.9	0.999

Stock concentrations follow Eq. (3), for example $10\ \mu\text{L}/20\ \text{mL} = 0.500\ \mu\text{L/mL}$. In the notebook the first set of readings and the relative dilution labels (1.0, 0.5, ...) were crossed out and replaced by the absolute concentrations and the final absorbances reported below; only the final values are used.

8.2 Standard-Curve Data

Table 3: Absorbance at $\lambda_{\max}$ for the five standards of each trial (c in µL/mL).

	Blue 1		Blue 2		Green		Yellow		Red	
Std	c	A	c	A	c	A	c	A	c	A
1	0.5	0.774	0.5	0.885	0.15	0.901	1.667	1.032	0.4	0.999
2	0.25	0.385	0.25	0.458	0.075	0.439	0.833	0.538	0.2	0.560
3	0.125	0.226	0.125	0.252	0.0375	0.239	0.417	0.269	0.1	0.308
4	0.0625	0.119	0.0625	0.140	0.01875	0.122	0.208	0.160	0.05	0.168
5	0.03125	0.062	0.03125	0.066	0.009375	0.061	0.104	0.084	0.025	0.099

8.3 Standard Curves

Fig. 2 shows the five standard curves with their least-squares lines. The sixth panel plots absorbance against the fraction of the stock concentration, which places all dyes on the same x-axis and shows that the two-fold dilution series behaved similarly for every dye.

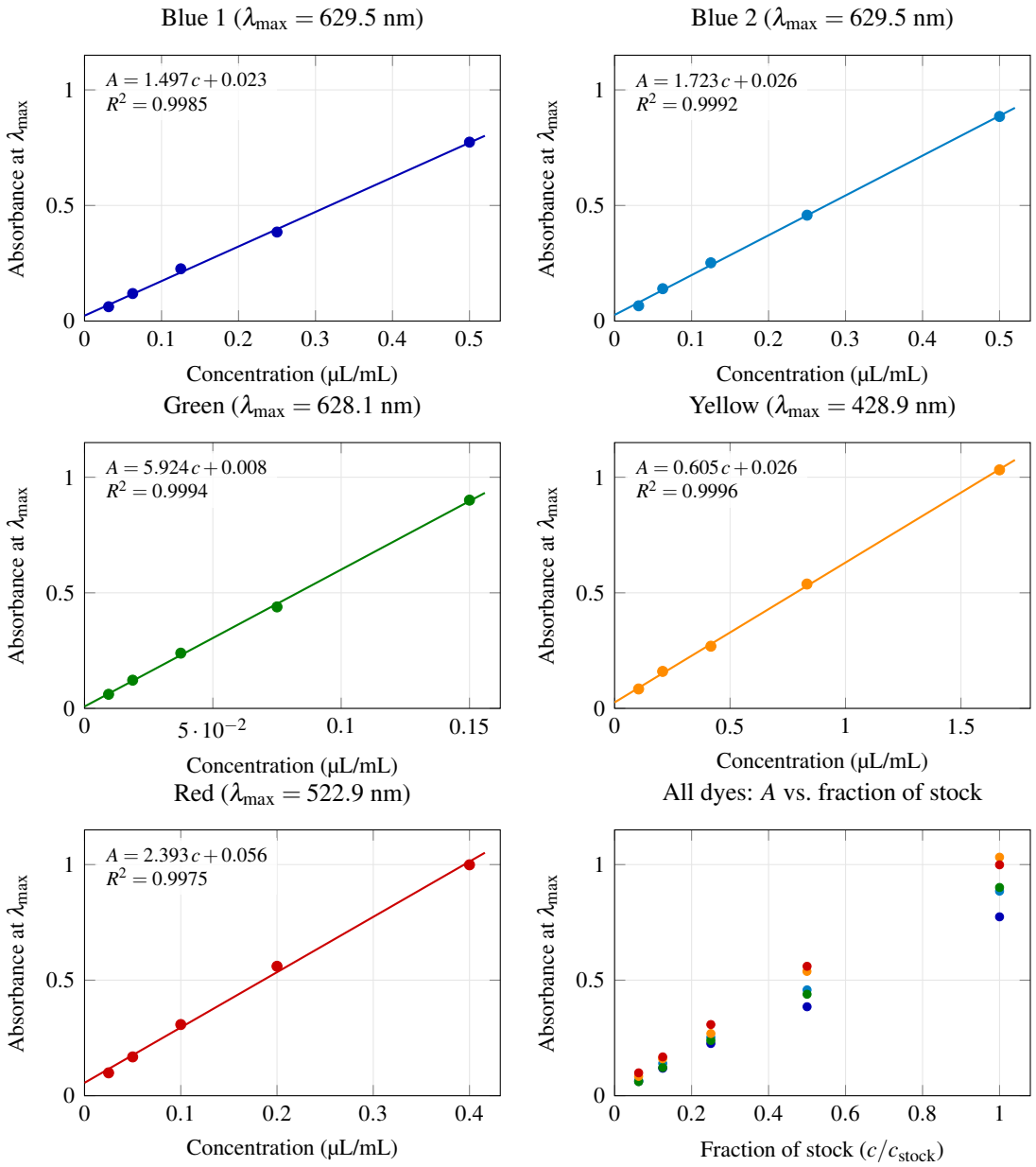


Figure 2: Standard curves at λ_{max} for blue (two trials), green, yellow and red, with linear fits and the normalised comparison. Concentration is the volume of undiluted dye per volume of solution.

8.4 Regression Results

Table 4: Linear regression $A = mc + b$ for each standard curve. Slopes and intercepts are given with 95 % confidence intervals ($n = 5$, 3 degrees of freedom). m and LOD/LOQ are in absorbance units per µL/mL and µL/mL, respectively.

Trial	$\lambda_{\max}$ (nm)	Slope m	Intercept b	R^2	$s_{y/x}$	LOD	LOQ
Blue 1	629.5	1.497 ± 0.105	0.023 ± 0.027	0.9985	0.0126	0.028	0.084
Blue 2	629.5	1.723 ± 0.090	0.026 ± 0.023	0.9992	0.0108	0.021	0.063
Green	628.1	5.924 ± 0.273	0.008 ± 0.021	0.9994	0.0098	0.005	0.017
Yellow	428.9	0.605 ± 0.023	0.026 ± 0.020	0.9996	0.0091	0.050	0.150
Red	522.9	2.393 ± 0.221	0.056 ± 0.046	0.9975	0.0212	0.029	0.089

8.5 Rubric Criteria

Table 5: Quantitative criteria from the Experiment 2 rubric.

Trial	$\lambda_{\max}$ (nm)	Max A	Max $A < 1$	R^2	$R^2 > 0.97$
Blue 1	629.5	0.774	✓	0.9985	✓
Blue 2	629.5	0.885	✓	0.9992	✓
Green	628.1	0.901	✓	0.9994	✓
Yellow	428.9	1.032	×	0.9996	✓
Red	522.9	0.999	✓	0.9975	✓

All five curves meet $R^2 > 0.97$. Four of five have a maximum absorbance below 1. The yellow top standard (1.032) is above 1 but inside the 0.9–1.1 target range stated in the lab manual. The red top standard (0.999) is just under the limit. Data quality, graph quality and instructor approval are assessed by the instructor and are not scored here.

9 Calculations

9.1 Stock and Standard Concentrations

$$c_{\text{blue 1}} = \frac{10\ \mu\text{L}}{20\ \text{mL}} = 0.500\ \mu\text{L/mL}, \quad c_{\text{yellow}} = \frac{20\ \mu\text{L}}{12\ \text{mL}} = 1.667\ \mu\text{L/mL}$$

$$c_{\text{yellow, std 2}} = \frac{1.667}{2} = 0.833, \quad c_{\text{std 3}} = 0.417, \quad c_{\text{std 4}} = 0.208, \quad c_{\text{std 5}} = 0.104\ \mu\text{L/mL}$$

9.2 Least-Squares Fit (Example: Yellow)

With $\bar{c} = 0.646$ µL/mL, $S_{xx} = \sum (c_i - \bar{c})^2 = 1.62$ and the residuals in Table 6, Eqs. (2)–(5) give

$$m = 0.6051 \pm 0.0228, \quad b = 0.0258 \pm 0.0196, \quad s_{y/x} = 0.0091,$$

$$\text{LOD} = \frac{3.3(0.0091)}{0.6051} = 0.050\ \mu\text{L/mL}, \qquad \text{LOQ} = \frac{10(0.0091)}{0.6051} = 0.150\ \mu\text{L/mL}.$$

The other trials are in Table 4.

9.3 Predicting a Better Yellow Stock

The fitted line predicts the absorbance of a more dilute yellow stock. For 18 µL in 12 mL ($c = 1.50$ µL/mL):

$$A = 0.6051(1.50) + 0.0258 = 0.93,$$

which would keep the top standard within 0.9–1.0.

9.4 Undiluted-Dye Concentration from the Slope

From the Beer's law interpretation of the slope, $m = \varepsilon \ell c_{\text{dye}} \times 10^{-3}$, so

$$c_{\text{dye}} = \frac{1000\, m}{\varepsilon\, \ell}. \tag{6}$$

For blue 1 with $\varepsilon = 130\,000$ L cm^{-1} mol^{-1} and $\ell = 1$ cm,

$$c_{\text{dye}} = \frac{1000(1.4965)}{130\,000} = 0.0115 \text{ M } (\approx 9.1 \text{ g/L with } M = 793 \text{ g/mol}).$$

This assumes a single pure dye with the literature ε at $\lambda_{\max}$; see Section 10.7.

10 Data Analysis

10.1 Linearity and Residuals

Fig. 3 and Table 6 show the residuals $A_i - \hat{A}_i$. All residuals are within ± 0.026 absorbance units (at most 2.6 % of the top standard), and no curve shows a gross outlier. Blue 1 has its largest residual at the middle standard ($+0.016$). The red residuals follow a curved pattern $(-, +, +, -, -)$, which suggests slight curvature or a systematic dilution error rather than random scatter.

Table 6: Residuals $A_i - \hat{A}_i$ (standards 1–5, highest to lowest concentration) and the intercept t-statistic b/s_b. The critical value is 3.18 ($\alpha = 0.05$, 3 d.f.).

Trial	1	2	3	4	5	t_b
Blue 1	+0.0025	-0.0124	+0.0157	+0.0022	-0.0080	2.73
Blue 2	-0.0029	+0.0009	+0.0103	+0.0059	-0.0142	3.60
Green	+0.0043	-0.0134	+0.0088	+0.0029	-0.0026	1.21
Yellow	-0.0025	+0.0081	-0.0092	+0.0083	-0.0048	4.20
Red	-0.0140	+0.0255	+0.0128	-0.0076	-0.0168	3.90

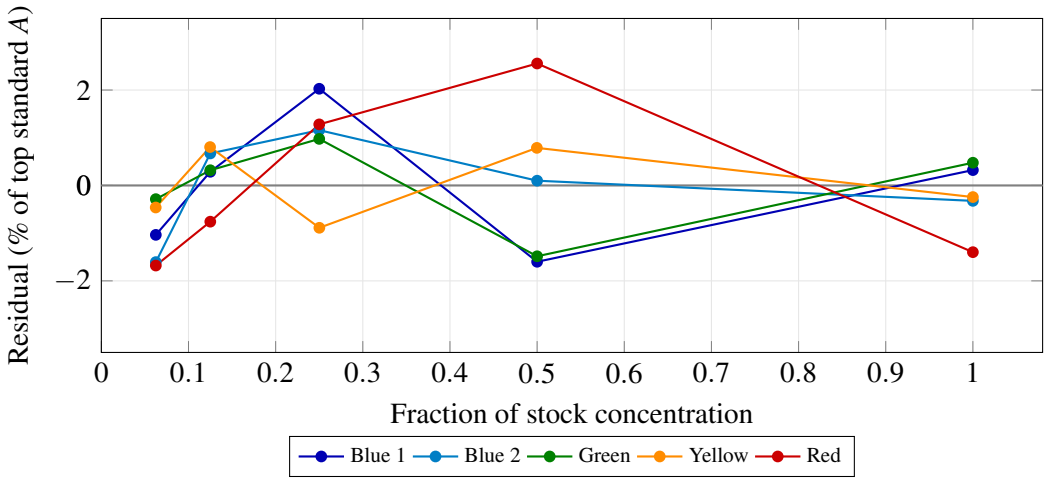


Figure 3: Residuals of the five standard curves, as a percentage of each curve's top standard absorbance. There is no common trend, but red shows a systematic pattern.

10.2 Intercepts and Serial-Dilution Consistency

Beer's law predicts a zero intercept, but all five intercepts are positive (0.008 to 0.056). The 95 % interval excludes zero for blue 2 ($b = 0.026 \pm 0.023$), yellow (0.026 ± 0.020) and red (0.056 ± 0.046), and includes zero for blue 1 and green (Table 6). A positive intercept means that the standards absorb slightly more than the blank, which is consistent with a small blanking offset, a slightly different cuvette position, or carry-over of dye in the dilution. It also explains why the ratio of absorbances of adjacent two-fold standards is below 2 (Table 7): for a pure proportional response it would be exactly 2.

Table 7: Ratio A_n/A_{n+1} of adjacent standards (expected 2.00 for an ideal two-fold dilution with zero intercept).

Trial	1/2	2/3	3/4	4/5	Mean
Blue 1	2.01	1.70	1.90	1.92	1.88
Blue 2	1.93	1.82	1.80	2.12	1.92
Green	2.05	1.84	1.96	2.00	1.96
Yellow	1.92	2.00	1.68	1.90	1.88
Red	1.78	1.82	1.83	1.70	1.78

10.3 Reproducibility: Blue 1 versus Blue 2

Both blue trials gave the same $\lambda_{\max}$ (629.5 nm) and linear curves ($R^2 = 0.9985$ and 0.9992). The slopes, however, differ: 1.497 ± 0.105 and 1.723 ± 0.090 (95 % CI). The difference is 0.226 µL/mL^{-1} (≈15 %), and with a combined standard error of 0.044 it gives $t \approx 5.2$, which is larger than the critical value for 3 d.f., so the trials are statistically different. Because both stocks were nominally 0.500 µL/mL (10 µL/20 mL and 5 µL/10 mL), the 15 % difference suggests that the stock was not made up as accurately as nominal: the 5 µL pipetting step is more sensitive to error than the 10 µL one (a 0.5 µL error is 10 % versus 5 %). Any unknown measured against a curve should use a curve from the same stock.

10.4 Sensitivity and Detection Limits

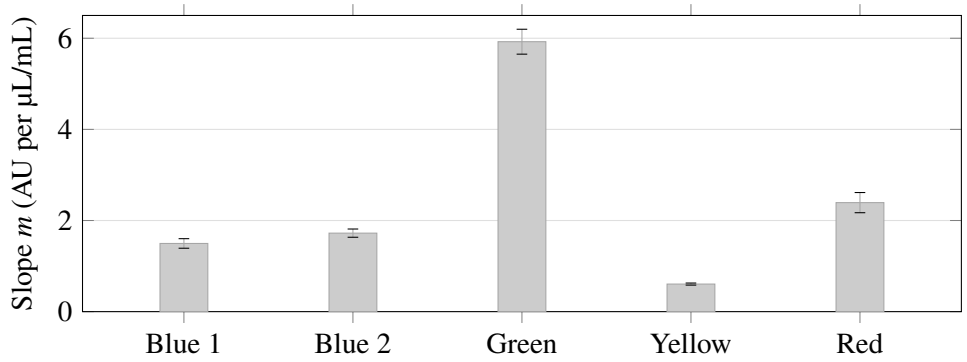


Figure 4: Standard-curve slopes with 95 % confidence intervals. The slope depends on how concentrated the undiluted dye is and on ε at $\lambda_{\max}$, not on the dye alone.

The slopes span a factor of about ten (yellow 0.605, blue 1.50–1.72, red 2.39, green 5.92 AU per µL/mL), reflecting both ε and the strength of each concentrated product, which is why a different stock concentration was needed for each dye to bring the top standard to $A \approx 1$. LOD estimates are 0.006–0.050 µL/mL and LOQ 0.017–0.150 µL/mL (Table 4). Expressed as a fraction of the top standard, the LOD is about 3–7 % for every dye, so the lowest standard of each series (6.25 % of the stock) lies close to the LOD, and below the LOQ, which is 9–22 % of the top standard.

10.5 Maximum Absorbance and the Linear Range

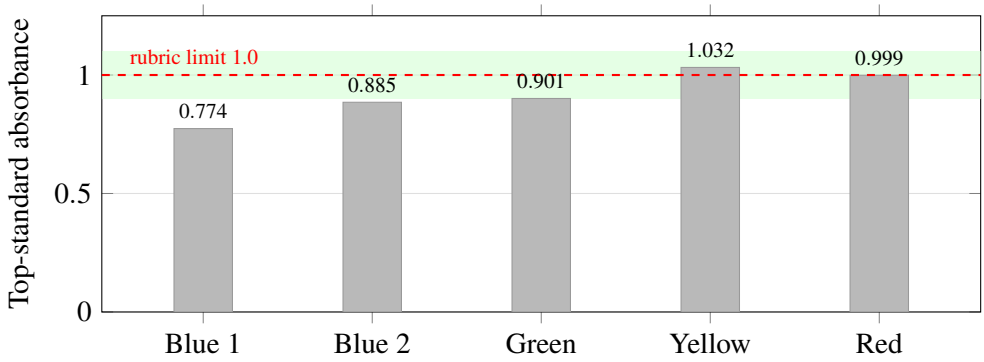


Figure 5: Top-standard absorbance of each curve. The shaded band is the 0.9–1.1 target range from the lab manual; the dashed line is the rubric limit of 1.

Blue 1 (0.774) is below the 0.9 target, and yellow is above 1.0. In neither case is the linearity affected: the highest standards of both lie on their fitted lines within 0.003.

10.6 Identification of the Dyes from λ_{max}

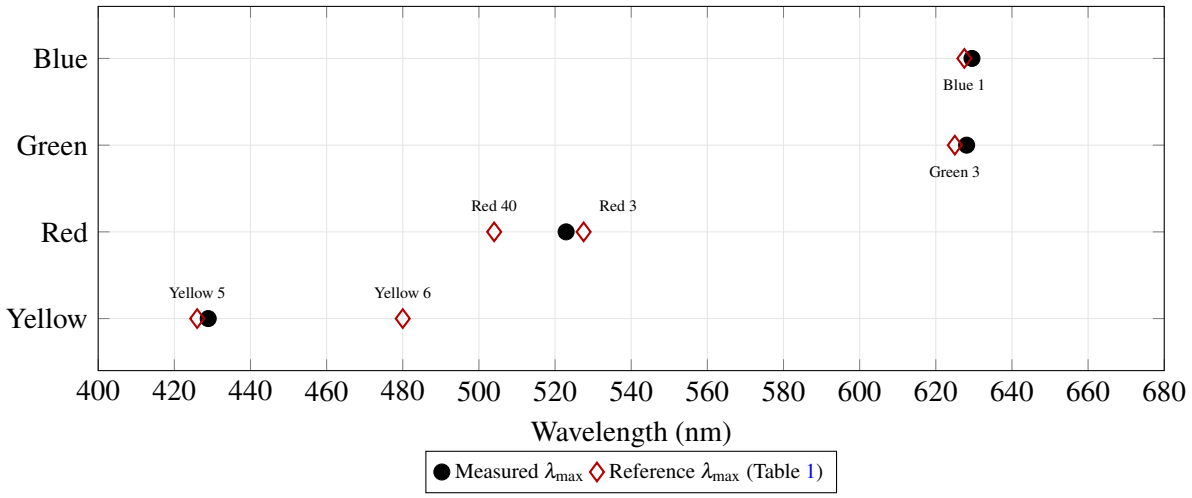


Figure 6: Measured λ_{max} (dots) compared with reference values (diamonds) for the dyes in Table 1. Measured values are within about 3 nm of Yellow 5 and Blue 1 and Green 3, and lie closest to Red 3 for red.

Yellow (428.9 nm) is within 2.9 nm of tartrazine (Yellow 5, 426 nm) and far from sunset yellow (480 nm). Blue (629.5 nm) falls inside the 625 nm–630 nm window of Brilliant Blue FCF. Red (522.9 nm) is ≈19 nm from Allura Red (504 nm) and 2–7 nm below erythrosine (525 nm–530 nm), so the absorbance maximum is consistent with erythrosine, or with a mixture. The green (628.1 nm) is within 3 nm of Fast Green FCF (625 nm) but also within 1.4 nm of the blue λ_{max}; because green food colorings are often prepared by blending a blue and a yellow dye, the band measured at 628 nm may come mainly from a blue component. A full spectrum of the green would show whether a second (yellow) band is present; the notebook records only the peak.

10.7 Undiluted-Dye Concentration (Consistency Check)

Table 8 applies Eq. (6), assuming that each coloring is a single pure dye with the literature ε.

Table 8: Estimated concentration of undiluted dye, assuming a single pure dye with literature ε (L cm^{-1} mol^{-1}) and $\ell = 1$ cm.

Trial	Assumed dye	ε	m	c_{dye} (mM)	g/L	% w/v
Blue 1	Blue No. 1	130 000	1.497	11.5	9.1	0.9
Blue 2	Blue No. 1	130 000	1.723	13.3	10.5	1.1
Yellow	Yellow No. 5	27 300	0.605	22.2	11.8	1.2
Red	Red No. 3	31 000	2.393	77.2	69.3	6.9
Green	Green No. 3	43 000	5.924	137.8	111.5	11.1

For blue (about 9–10 g/L, 1 % w/v) and yellow (about 12 g/L) the values are of a plausible order for a liquid food coloring. The red (about 69 g/L) and green (about 110 g/L) values are very high and are unlikely to be real dye contents. This is expected if the colorings are mixtures or if the dye has a different absorptivity from

the pure-dye reference, so these two values are not used further. The check is only an order-of-magnitude one.

10.8 Note on the Spreadsheet Graphs

The trendline equations shown on the exported spreadsheet graphs (e.g. $y = 0.6676x - 0.0153$ for blue 1 and $y = 0.1687x - 0.0013$ for green) are the reciprocal of the slopes in Table 4 ($1/1.4965 = 0.668$, $1/5.924 = 0.169$): the graphs were fitted with concentration as the dependent variable. The R^2 values are the same either way, but the equation for a standard curve should be $A = mc + b$ (absorbance on the y-axis, as stated in the manual), so the slopes in this report should be used for calibration.

11 Discussion

11.1 Interpretation

Hypothesis 1 is supported: each dye gave one dominant visible maximum at a wavelength consistent with its color, and the measured λ_{max} agree with the reference dyes in Table 1 within a few nm (blue, yellow), or point to a mixture or a different dye (red, green). Hypothesis 2 is supported: all five calibration lines have $R^2 \geq 0.9975$ and residuals within 2.6 %. Hypothesis 3 is not supported: the two blue slopes differ by about 15 %, a statistically significant amount, and this is attributed to preparation of the stock rather than to the spectrophotometer, since λ_{max} and linearity were the same.

The positive intercepts (up to 0.056 for red, 5.6 % of the top standard) show that a small constant absorbance offset is present. It should be fitted rather than forced through zero, since forcing it would bias the slope (for blue 2 by about 4 %, 1.80 versus 1.72).

11.2 What Worked and What Did Not

Worked well:

- Two-fold serial dilution gave five evenly spaced standards covering a factor of 16 in concentration, with $R^2 > 0.997$ for every dye.
- Blanking with distilled water and measuring at λ_{max} gave the same λ_{max} for both blue trials.
- Preparing each dye at a different stock concentration brought every top standard to $A \approx 0.77$–1.03.

Did not work as well:

- The yellow top standard was above 1 (1.032), and blue 1 was below 0.9 (0.774). A more dilute yellow stock (about 18 µL in 12 mL) would give $A \approx 0.93$.
- Small-volume pipetting (3–8 µL) of the concentrated dye is the largest source of error and probably produced the blue 1/blue 2 difference.
- Several first readings and relative dilution labels were crossed out and replaced in the notebook, and the full spectra were not saved. Recording the spectra and the exact dilution volumes would let the analysis be checked.

- Each standard was measured once, so no replicate uncertainty is available; LOD and LOQ come from the scatter around the line only.

11.3 Sources of Error

(1) Micropipetting of 3 µL–20 µL of viscous dye (volume error, drops left in the tip). (2) Imperfect mixing or carry-over in the serial dilution, which accumulates down the series. (3) Cuvette scratches, fingerprints, bubbles and orientation (the cuvette was wiped and aligned for every reading). (4) Baseline offset after blanking. (5) Reading at the peak of a broad band is not sensitive to a 1–2 nm wavelength error, so wavelength setting is a minor error.

12 Conclusion

λ_{max} values of 629.5 nm (blue), 628.1 nm (green), 428.9 nm (yellow) and 522.9 nm (red) were determined, and standard curves of absorbance versus concentration were prepared for each dye. All curves were linear ($R^2 = 0.9975$–0.9996) and gave sensitivities of 0.605 (yellow), 1.50 and 1.72 (blue), 2.39 (red) and 5.92 (green) AU per µL/mL, with limits of quantitation of 0.017–0.150 µL/mL. Four of the five curves had a maximum absorbance below 1 (yellow: 1.032). The two blue slopes differ by about 15 %, so a calibration curve should always be prepared from the same stock as the sample. These curves can be used to determine the concentration of the same food coloring in an unknown beverage sample.

References

1. Chemistry 250L, Experiment 2: *Spectrophotometers and Standard Curves*. Lab manual, Brigham Young University–Idaho.
2. Chemistry 250L, *The Chemistry of Food Coloring*. Course reading.
3. Chemistry 250L, *Experiment 2 Rubric*. Course document.

A Raw Data (laboratory notebook)

Values transcribed from the notebook (final values; crossed-out earlier readings omitted): dye volume / final volume, λ_{max} (nm), absorbance of the standards from highest to lowest concentration.

- Blue 1: 10 µL/20 mL, 629.5; 0.774, 0.385, 0.226, 0.119, 0.062.
- Blue 2: 5 µL/10 mL, 629.5; 0.885, 0.458, 0.252, 0.140, 0.066.
- Green: 3 µL/20 mL, 628.1; 0.901, 0.439, 0.239, 0.122, 0.061.
- Yellow (orange): 20 µL/12 mL, 428.9; 1.032, 0.538, 0.269, 0.160, 0.084.
- Red: 8 µL/20 mL, 522.9; 0.999, 0.560, 0.308, 0.168, 0.099.